



COMP 385 001

College of Engineering
Department of Computer Science
Fall 2026 Course Syllabus
Theory of Computing

NOTE: Students are responsible for reading, understanding, and following the syllabus.

Course Information

Class Location: GIBBS 337

Meeting Times: 1530-1645 TR

Instructor Contact Information

Letu Qingge

Email: lqingge@ncat.edu

Office Location: Martin 327

Office Phone: 3362853754

Student Hours

Thursday 2PM-3PM

Course Prerequisites

Prerequisites: COMP360

Course Description

This course is the study of topics that include the theory of finite state machines and automata; regular expressions; Turing machines; grammars; parsing; the language hierarchy; machine design and construction; computability; insolvability; and the halting problem. The course also discusses issues in the equivalence of various computational models and characterizations. Prerequisites: COMP 360.(F;S;SS)

Student Learning Objectives/Outcomes (SLO)

Upon the completion of this course:

1. Apply computer science theory and software development fundamentals to produce computing-based solutions.

Career Competencies

Analyze a problem and define the computing requirements for solution

- Student has ability to design and manipulate Automata for Languages.

Reason about and explain computer-based solutions at levels of abstraction

- Student has ability to explain and manipulate what a Turing Machine and Bayes theorem for solving problems.
- Student has ability to define the classes P, NP, and NP-complete.

Required Textbooks and Materials

Introduction to the Theory of Computation

Authors: Michael Sipser,

Edition: Third Edition

Grading Scale

Course Grading Scale

Letter Grade	Numerical
A	94 and above
A-	90-93
B+	87-89
B	84-86
B-	80-83
C+	77-79
C	74-76
C-	70-73
D+	65-69
D	60-64
F	0-59

Grading Allocation

Grading Allocation

Assignments	Percentage or Points
Assignments	30%
Class Participation	10%
Midterm Exam	30%
Final Exam	30%
Total	100

Use Of Canvas as The Learning Management System

Canvas is the primary online instructional and course communications platform. Students can access the course syllabus, assignments, grades, and learner support resources. Students are encouraged to protect their login credentials, complete a Canvas training, and log in daily to the course.

Make-Up Exams

N/A

Extra Credit

N/A

Late Work

Late Work will NOT be accepted.

Late submissions will result in a grade of zero for any assignments. No late work is allowed unless a special reason is presented with documentation required such as sickness; death of relatives (immediate family); participation in approved University related activities; acting in the capacity of a representative of the University (band, choir, sports related travel, etc.); and extraordinary circumstances (court appearance, family emergency, etc.). NOTE: Other reasons are not acceptable. You are not allowed to have others to turn in assignment for you. However, any late submission because of your special reasons is not accepted after a week from the deadline.

Special Assignments

N/A

AI Course Policy

Generative AI is prohibited on assignments and exams.

Class Schedule

Date	Subject	Reading, Activity, Homework, Exam
8/20	Course Overview, Basics Concepts	
8/25	Mathematical Preliminaries and Notation, Basic Concepts	
8/27	Deterministic finite automata (DFA)	HW1
9/1	Regular Languages	
9/3	Non-deterministic finite automata (NFA)	
9/8	Pumping Lemma	HW2
9/10	Pumping Lemma	
9/15	Context-free Languages	
9/17	Context-free Languages	HW3
9/22	PDA=CFL	
9/24	Decidability	
9/29	Decidability	HW4
10/1	Undecidability	
10/6	Midterm Exam	In-class exam
10/8	Undecidability	
10/15	Reducibility	
10/20	Reducibility	
10/22	Time Complexity	HW5
10/27	Dynamic Programming	
10/29	Dynamic Programming	
11/3	Class NP	HW6
11/5	Class NP	

Date	Subject	Reading, Activity, Homework, Exam
11/12	NP-C Problems	
11/17	NP-C Problems	HW7
11/19	Polynomial Reduction	
11/24	Polynomial Reduction	
12/1	Solve NP-C Problems	HW8
12/3	Solve NP-C Problems	
12/10	Final Exam	2 pm – 4 pm

Additional Course Information

None

University Policies

Aggies-Care Link

<https://www.ncat.edu/coronavirus/aggies-care/index.php>

AI Ethical Use First Steps: The following language is intended to guide the use of AI in teaching and learning.

When incorporating ChatGPT or any other generative artificial intelligence (AI) technology into course materials, assignments, or discussions, faculty members at the university should promote ethical use of these tools. This includes providing proper attribution when AI-generated content is used or presented. Citations should follow the guidelines specified by academic writing styles such as APA, IEEE, MLA, or other relevant citation styles. Here's an example from APA:

<https://apastyle.apa.org/blog/how-to-cite-chatgpt>

While faculty members may not be able to always individually verify the originality of all student work in both teaching and research settings, they can play a crucial role with respect to educating student about the importance of originality in their work and upholding university values related to academic integrity. Faculty members are encouraged to explore alternative assessment approaches and technology tools that promote academic integrity and facilitate the identification of AI-generated content when necessary.

It is essential for faculty members and students to be aware of the licensing agreements and terms of service associated with the use of AI language models like ChatGPT or other generative AI technologies. Different language models may have varying usage restrictions, and compliance with ethical and legal guidelines is crucial. Faculty members are encouraged to refer to official documentation provided by the language model's developers, such as OpenAI's website, and any relevant platform-specific guidelines with respect to language model or other generative AI technologies, such as for ChatGPT usage. Standards specific to the dissemination of scholarly products in various venues are likewise essential (conferences, journals, etc).

Guidance for the syllabus:

Regarding language model and other generative AI technologies, including those such as ChatGPT, and their relationship to teaching and learning, the policy in the syllabus should be explicitly stated with reference to the university [Academic Dishonesty Policy](#). Faculty members have the flexibility to choose from the following options if relevant to their course of instruction:

- **Not Permitted:** Assistance from large language model and other generative AI technologies, including those such as ChatGPT, is not allowed for any activities or assignments. All submissions must be the original work of the student, and the use of AI-generated content is strictly prohibited. Possible disciplinary action for violations of this course policy is described in the university's Academic Dishonesty Policy.
- **Permitted for Some Activities with Attribution:** The use of large language model and other generative AI technologies, including those such as ChatGPT, is allowed only for specified activities or assignments, but proper attribution is required. Students should clearly indicate when AI-generated content has been used and cite the specific language model and platform employed following the citation guidelines specified by APA, MLA, or any other relevant academic style. However, the majority of the work must still be the original creation of the student. Possible disciplinary action for violations of this course policy is described in the university's Academic Dishonesty Policy.

Permitted for All Activities with Attribution: The use of large language model and other generative AI technologies, including those such as ChatGPT, is permitted for all activities and assignments, but proper attribution is required. Students should acknowledge the use of AI-generated content and provide appropriate citations following the guidelines specified by APA, MLA, or any other relevant academic style. It is important to emphasize that even with AI assistance, students are expected to engage critically and creatively with the material to produce their work. Possible

disciplinary action for violations of this course policy is described in the university's Academic Dishonesty Policy.

Participation

Students are expected to attend class and participate on a regular basis in order to successfully achieve course learning outcomes and meet federal financial aid requirements ([34 CFR 668.22](#)). Active participation may consist of course interactions with the content, classmates, and/or the instructor. Examples of academically-related course activities include, but are not limited to:

- Completing and submitting assignments, quizzes, exams, and other activities within Canvas or through Canvas (3rd-party products).
- Participating in course-related synchronous online chats, discussions, or meeting platforms such as Zoom in which participation is tracked.

Note: Uploading assignments through Canvas presents a challenge for Chromebook users in locating the files for submission. If you use a Chromebook, please be sure you also have access to a Mac computer or Windows computer so you can fully participate in your Canvas class. For more information about student computer recommendations, please visit <https://hub.ncat.edu/administration/its/computer-recommendations.php>.

Classroom Citizenship

Courtesy, civility, and respect must be the hallmark of your interactions. It is expected that students will conduct themselves in a courteous manner to fellow students and the instructor.

Cell Phone Usage

Cell phone use inside classrooms during class periods is prohibited. Be advised that placing or receiving calls, as well as conversing on cell phones during class, shall be considered as disruptive behavior for students and may be subject to the procedures on Disruptive Students in the Classroom.

Compliance with the Americans with Disabilities Act

North Carolina A&T State University (N.C. A&T) is committed to following the requirements of the Americans with Disabilities Act Amendments Act (ADAAA) and Section 504 of the Rehabilitation Act. Suppose you need an academic accommodation based on the impact of a disability. In that case, you must initiate the request with the Office of Accessibility Resources (OARS) and provide documentation in accordance with the Documentation Guidelines at N.C. A&T. Once documentation is received, it will be reviewed. Once approved, you must attend a comprehensive meeting to receive appropriate and reasonable accommodations. If you are a student registered with OARS, you must complete the Accommodation Request Form to have accommodations sent to faculty.

OARS is located in the Academic Classroom Building, Room 302 and can be reached at 336-334-7765, or by email at accessibilityresources@ncat.edu. Additional information and forms can be found on the internet at <https://www.ncat.edu/provost/academic-affairs/accessibility-resources/index.php>.

Please note: Accommodations are not retroactive and begin once the Disability Verification Form is provided to faculty.

Title IX

North Carolina A&T State University is committed to providing a safe learning environment for all students—free of all forms of discrimination and harassment. Sexual misconduct and relationship violence in any form are inconsistent with the university's mission and core values, violate university policies, and may also violate federal and state law. Faculty members are considered "Responsible Employees" and are required to report incidents of sexual misconduct and relationship violence to the Title IX Coordinator. If you or someone you know has been impacted by sexual harassment, sexual assault, dating or domestic violence, or stalking, please visit the Title IX website to access information about university support and resources <https://www.ncat.edu/legal/title-ix/index.php>. If you would like to

speak with someone confidentially, please contact the Counseling Services at 336-334-7727 or the Student Health Center at 336-334-7880.

Technical Requirements

Students are to refer to the Information Technology (IT) Services student computer requirement list to determine the minimum technical requirements based on College specifications. Link:

<https://hub.ncat.edu/administration/its/computer-recommendations.php>

Online Orientation Requirement

All new students at North Carolina Agricultural and Technical State University are required to complete an online orientation, which includes familiarizing themselves with the LMS tools (Canvas) and online learning strategies before starting their first course.

<https://bit.ly/AggieOnlineLearningOrientation>

How to Enroll

<https://bit.ly/AggieOnlineLearningOrientationSelfEnroll>

Technical Support

If you experience any problems with your N.C. A&T account, you may call Client Technology Services (formerly Aggie Tech Support and Help Desk) at 336-334-7195, or visit

<https://hub.ncat.edu/administration/its/dept/ats/index.php>

Field Trip Policies / Off-Campus Instruction and Course Activities

Off-campus, out-of-state, and foreign instruction and activities are subject to state law and university policies and procedures regarding travel and risk-related activities. Information regarding these rules and regulations may be found at <https://www.ncat.edu/campus-life/student-affairs/index.php>

Student Handbook

<https://www.ncat.edu/campus-life/student-affairs/departments/dean-of-students/student-handbook.php>

Undergraduate Catalog

<https://catalog.ncat.edu/>

Student Travel Procedures and Student Travel Activity Waiver

https://hub.ncat.edu/administration/student-affairs/staff-resources/student_activity_travel_waiver.pdf

Sexual Misconduct Policy (Title IX website)

<https://www.ncat.edu/legal/title-ix/policies.php>

Family Educational Rights and Privacy Act (FERPA)

<https://www.ncat.edu/registrar/ferpa.php>

Grade Appeal

<https://catalog.ncat.edu/content.php?catoid=16&navoid=616#grade-appeal>

Student Complaint Policy

<https://hub.ncat.edu/administration/legal/policies/sec2-acad-affairs/student-complaint-policy-final.pdf>

Student Complaint Procedures

<https://www.ncat.edu/current-students/student-complaint-form.php>

Student Conduct and Discipline

North Carolina A&T State University has rules and regulations that govern student conduct and discipline meant to ensure the orderly and efficient

conduct of the educational enterprise. It is the responsibility of each student to be knowledgeable about these rules and regulations.

Please consult the following about specific policies such as academic dishonesty, cell phones, change of grade, disability services, disruptive behavior, general class attendance, grade appeal, incomplete grades, make-up work, student grievance procedures, withdrawal, etc.:

- Undergraduate Bulletin
<https://www.ncat.edu/provost/academic-affairs/bulletins/index.php>
- Graduate Catalog
<https://www.ncat.edu/tgc/graduate-catalog/index.php>
- Student Handbook
<https://www.ncat.edu/campus-life/student-affairs/departments/dean-of-students/student-handbook.php>

Academic Dishonesty Policy

Academic dishonesty includes but is not limited to the following:

1. Cheating or knowingly assisting another student in committing an act of cheating or other academic dishonesty;
2. Plagiarism (unauthorized use of another's words or ideas as one's own), which includes but is not limited to submitting exams, theses, reports, drawings, laboratory notes or other materials as one's own work when such work has been prepared by or copied from another person;
3. Unauthorized possession of exams or reserved library materials; destroying or hiding source, library or laboratory materials or experiments or any other similar actions;
4. Unauthorized changing of grades, or marking on an exam or in an instructor's grade book or such change of any grade record;
5. Aiding or abetting in the infraction of any of the provisions anticipated under the general standards of student conduct;

6. Hacking into a computer and gaining access to a test or answer key prior to the test being given. N.C. A&T reserves the right to search the emails and computers of any student suspected of such computer hacking if a police report of the suspected hacking was submitted prior to the search; and
7. Assisting another student in violating any of the above rules.

A student who has committed an act of academic dishonesty has failed to meet a basic requirement of satisfactory academic performance. Thus, academic dishonesty is not only a basis for disciplinary action but may also affect the evaluation of a student's level of performance. Any student who commits an act of academic dishonesty is subject to disciplinary action.

In instances where a student has clearly been identified as having committed an act of academic dishonesty, an instructor may take appropriate disciplinary action, including a loss of credit for an assignment, exam, or project; or awarding a grade of "F" for the course, **subject to review and endorsement by the chairperson and dean.**

The Office of Accessibility Resources

The Office of Accessibility Resources (OARS) works collaboratively with students, faculty, and staff to create a universally accessible campus where programs and services are usable by all people, to the greatest extent possible. OARS provides the campus community with resources and services for individuals with disabilities to achieve social justice and equity. New and returning students who request accommodations due to a disability or medical condition are responsible for registering with OARS, submitting documentation, knowing and following the policies and procedures of OARS located on our [Get Connected](#) webpage. OARS is located in the Academic Classroom Building, Room 302.

Phone: 336-334-7765

Email: accessibilityresources@ncat.edu

Office of Veteran & Military Affairs

The Office of Veteran and Military Affairs (OVMA) is here to meet the needs of our students who are veterans, service members, dependents of service members, and/or veterans. The OVMA certifies your educational benefits throughout each academic year. The OVMA is also prepared to advise and counsel you in regard to any veteran or education-related concerns. The OVMA serves as the liaison between veteran

students and the Department of Veterans Affairs. It assists eligible veteran students in understanding their educational benefits, provide guidance on procedural requirements, and certify, register, and report enrollments to the VA to ensure veterans receive their benefits in a timely manner.

The OVMA is located in The Aggie Student V.E.T.S. Center at The Oaks.
Phone: 336-398-8651

Counseling Services

Counseling Services, an [International Accreditation of Counseling Services](#) (IACS) accredited unit in the [Division of Student Affairs](#), offers a variety of services to help all currently enrolled North Carolina A&T students address challenges and difficulties they may face. Services are designed to help students understand themselves better, create and maintain healthy relationships, improve their academic performance, and make satisfying career and life choices. If you are experiencing distress, they provide a safe and confidential environment for counseling. You may contact Counseling Services at 336-334-7727 or stop by our office in Murphy Hall, Suite 109.

Tutorial Services

The mission of CAE Tutorial Program is to empower students for academic success within a safe and inclusive learning environment. Our goal is to provide academic support by offering a broad range of tutoring services for those students desiring to improve their in-class performance. When you schedule a tutoring session, you can get help with preparing for an upcoming exam or with covering difficult concepts you want to master.

North Carolina A&T State University has partnered with Upswing to help students find, schedule, and book tutoring sessions both in-person and virtually. Through a web-based tutoring platform, you can schedule FREE tutoring lessons with Upswing's online tutors, or with North Carolina A&T State University in-person tutors, for a variety of subjects. Go to ncat.upswing.io and log in with your Aggie OneID credentials to get started, or contact the Center for Academic Excellence at 336-334-7855.

Technology Support

Client Technology Services

Client Technology Services (CTS) is a part of Information Technology Services (ITS). This unit is responsible for desktop computing support and related software programs for the university. Support is provided in the academic labs, residential labs, and

administrative offices.

Phone: 336-334-7195

Email: helpdesk@ncat.edu

Canvas Log-In

<https://ncat.instructure.com/login/saml>

Log-in using your OneID credentials

For assistance with OneID accounts, contact Client Technology Services at 336-334-7195.

Academic Engagement

This course engages a range of scholarly perspectives to foster critical thinking, analysis, and debate. Inclusion of a reading does not imply endorsement of its views.

Useful Links

AGGIES CARE: N.C. A&T Protocols	https://www.ncat.edu/coronavirus/aggies-care/index.php
Student Computer Recommendations	https://hub.ncat.edu/administration/its/computer-recommendations.php
Accessibility Resource Email	https://www.ncat.edu/provost/academic-affairs/accessibility-resources/index.php
Office of Accessibility Resources	https://www.ncat.edu/provost/academic-affairs/accessibility-resources/index.php

Title IX	https://www.ncat.edu/legal/title-ix/index.php
Client Technology Services	https://hub.ncat.edu/administration/its/dept/ats/index.php
Division of Student Affairs	https://www.ncat.edu/campus-life/student-affairs/index.php
Student Handbooks	https://www.ncat.edu/campus-life/student-affairs/departments/dean-of-students/student-handbook.php
Graduate Catalog	https://www.ncat.edu/tgc/graduate-catalog/index.php
Student Activity Travel Waiver	https://hub.ncat.edu/administration/student-affairs/staff-resources/studen_activity_travel_waiver.pdf
Title IX Policies	https://www.ncat.edu/legal/title-ix/policies.php
FERPA	https://www.ncat.edu/registrar/ferpa.php

Student Complaint Procedure	https://www.ncat.edu/current-students/student-complaint-form.php
Undergraduate Bulletins	https://www.ncat.edu/provost/academic-affairs/bulletins/index.php